

4.6 and INET 2.5. It can be installed on all versions of Ubuntu and on MAC platforms.

The features of iCanCloud are listed below:

- Both existing and non-existing cloud computing architectures can be modelled and simulated.
- A flexible cloud hypervisor module provides an easy method for integrating and testing both new and existent cloud brokering policies.
- Customisable VMs can be used to quickly simulate uni-core/multi-core systems.
- iCanCloud provides a wide range of configurations for storage systems, which include models for local storage systems, remote storage systems like NFS, and parallel storage systems (like parallel file systems and RAID systems).
- iCanCloud provides a user-friendly GUI that makes it easier to generate and customise large distributed models. This GUI is especially useful for managing a repository of preconfigured VMs, a repository of preconfigured cloud systems, and a repository of preconfigured experiments to launch experiments from the GUI and generate graphical reports.
- iCanCloud provides a POSIX-based API and an adopted MPI library for modelling and simulating applications. Also, several methods for modelling applications can be used in iCanCloud—using traces of real applications, using a state graph, and programming new applications directly in the simulation platform.
- New components can be added to the repository of iCanCloud to increase its functionality.

EMUSIM

EMUSIM stands for Integrated Emulation and Simulation. It combines emulation (AEF—Automated Emulation Framework) and Simulation (CloudSim) to enable more accurate models of software artefacts to be used during simulations. EMUSIM was developed by Rodrigo N. Calherios at the Cloud Computing and Distributed Systems (CLOUDS) Laboratory, Department of Computing and Information Systems, University of Melbourne, Australia.

EMUSIM automatically extracts information about application behaviour via emulation and then uses this information to generate a corresponding simulation model.

Comparing open source cloud computing simulators

Simulator	Platform	Type	Programming language	Cost modelling	GUI	Communication model	Simulation time
CloudSim	SimJava	Open source	Java	Yes	No	Limited	Second
CloudAnalyst	CloudSim	Open source	Java	Yes	Yes	Limited	Second
GreenCloud	NS-2	Open source	C++, oTCL	No	Limited	Full	Minute
iCanCloud	SIMCAN	Open source	C++	Yes	Yes	Full	Second
EMUSIM	CloudSim, AEF	Open source	Java	Yes	No	Limited	Second
GroudSim	-	Open source	Java	No	Limited	No	Second
DCSim	-	Open source	Java	No	Limited	No	Second

The EMUSIM simulator is of great use when the tester has no idea about the performance of the software under the varied levels of concurrency and parallelism which impede simulation. These can replace *in-site* experiments that would require infrastructure that is either unavailable for the tester or too expensive to run in the public cloud. EMUSIM is open source software under the GPL License.

The latest version of AEF is 1.3, released in August 2010.

The features of EMUSIM are listed below:

- Offers a combination of simulation and emulation to evaluate the effect of varying resources and patterns of requests on cloud applications.
- Accurately models applications to supply information regarding performance.
- Reduces the cost of running cloud based simulation, because rather than local and limited infrastructure usage, a pay-as-you-go public cloud is used for evaluation purposes.
- Supports loosely coupled CPU-intensive applications.

GroudSim

GroudSim (Gr-Grid oud-Cloud) is an event-based simulator designed for scientific applications on grid and cloud environments. It only requires one simulation thread. GroudSim was developed by S. Ostermann, K. Plankensteiner and D. Bodner, and can also be termed as a grid and cloud simulation tool kit for scientific applications based on a scalable simulation-independent discrete-event core.

GroudSim provides a comprehensive set of features for complex simulation scenarios, ranging from simple job executions on leased computing resources to calculation of costs, and background load on resources. GroudSim mainly focuses on the IaaS area of cloud computing. It can be easily extended to additional models like SaaS and PaaS in cloud computing.

SimEngine is the main class of GroudSim, which implements the time advance algorithm, the clock and the future events list, keeping track of the registered entities used for tracing during a simulation. The grid and cloud resources classes share most of the common functionality implemented in the Groud package and override the specialised behaviour in the Groud.